

ClearPath Database Schema

Document: 09 of 10 Phase: Phase 3 Status: Complete Generated: 2026-04-04

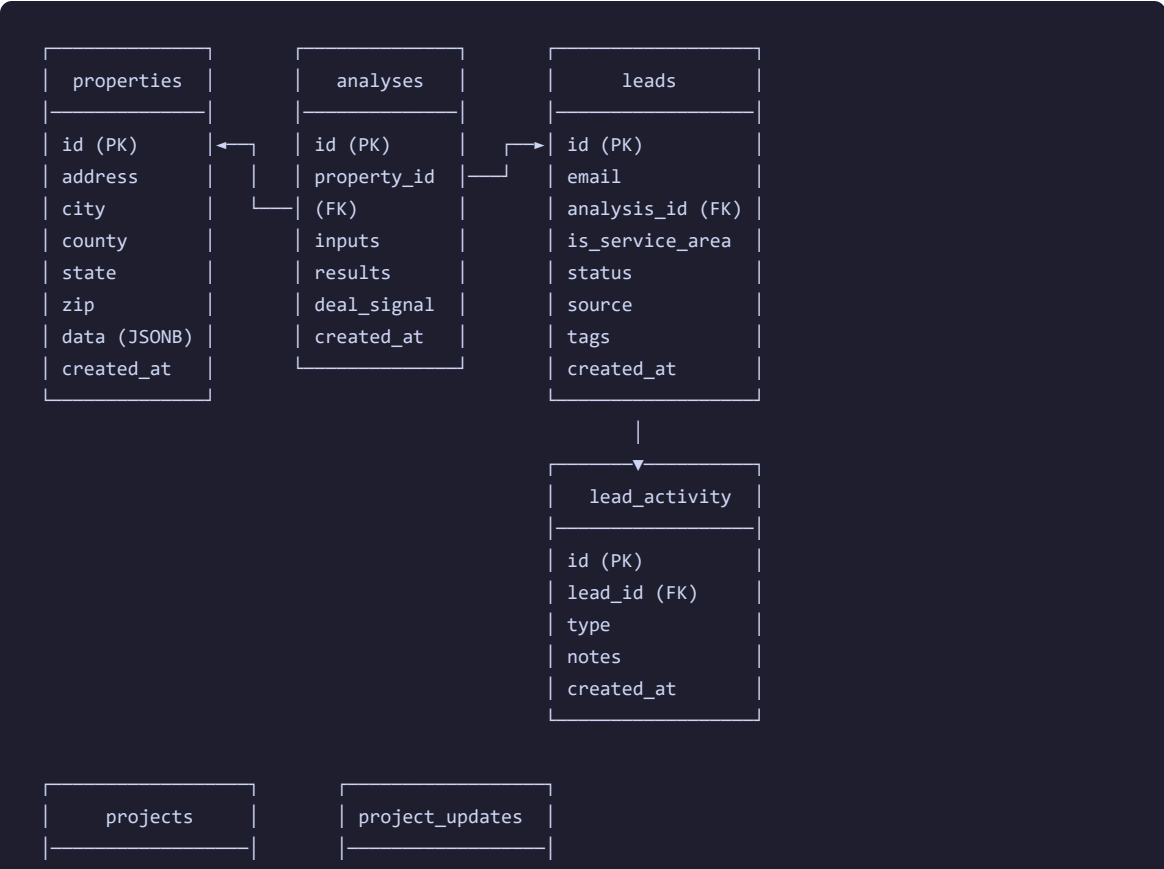
Overview

The ClearPath database is hosted on **Supabase (PostgreSQL)**. It is designed to be simple, flat where possible, and JSONB-flexible where the data shape may evolve. Every table has Row Level Security (RLS) enabled.

Design Principles:

- No premature normalization — keep it flat until complexity demands otherwise
- Use JSONB for variable-shape data (analysis inputs/outputs, property data)
- Timestamps on everything
- Soft deletes where applicable (no data destruction)

Entity Relationship Diagram



id (PK)	←	project_id (FK)
lead_id (FK)		id (PK)
property_id (FK)		type
status		content
sow (JSONB)		photos
budget		created_at
created_at		

Table Definitions

1. `properties`

Stores raw property data pulled from Zillow/API or entered manually. One row per unique address.

```
CREATE TABLE properties (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  address     TEXT NOT NULL,
  city        TEXT,
  county      TEXT,          -- "Cook" for service area detection
  state       TEXT DEFAULT 'IL',
  zip         TEXT,
  lat         DECIMAL(10, 7),
  lng         DECIMAL(10, 7),
  sqft        INTEGER,
  bedrooms    INTEGER,
  bathrooms   DECIMAL(3, 1), -- 2.5 baths
  year_built  INTEGER,
  lot_size_sqft INTEGER,
  property_type TEXT,        -- 'single_family', 'multi_family', 'condo'
  zestimate   INTEGER,       -- Zillow estimate at time of pull
  data        JSONB,         -- Full raw API response for future use
  is_service_area BOOLEAN GENERATED ALWAYS AS (county = 'Cook') STORED,
  created_at  TIMESTAMPTZ DEFAULT now(),
  updated_at  TIMESTAMPTZ DEFAULT now()
);

-- Indexes
CREATE INDEX idx_properties_address ON properties (address);
CREATE INDEX idx_properties_zip ON properties (zip);
CREATE INDEX idx_properties_county ON properties (county);
CREATE INDEX idx_properties_service_area ON properties (is_service_area);
```

Key design decision: `is_service_area` is a **generated column** — it auto-computes from `county = 'Cook'`. No need to set it manually. If the service area ever expands, update this one expression.

2. analyses

One row per deal analysis run. A single property can have multiple analyses (different conditions, prices).

```
CREATE TABLE analyses (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  property_id UUID REFERENCES properties(id) ON DELETE SET NULL,

  -- User inputs (stored explicitly for auditability)
  input_address TEXT NOT NULL,
  input_condition TEXT NOT NULL CHECK (input_condition IN (
    'cosmetic', 'light', 'medium', 'heavy', 'gut'
  )),
  input_purchase_price INTEGER NOT NULL,
  input_down_pct    DECIMAL(4, 3) DEFAULT 0.25,
  input_hold_months INTEGER DEFAULT 6,
  input_interest_rate DECIMAL(5, 4) DEFAULT 0.075,

  -- All inputs as JSONB (for flexibility / future fields)
  inputs          JSONB,

  -- Calculated results
  arv             INTEGER,
  rehab_low       INTEGER,
  rehab_high      INTEGER,
  rehab_estimate  INTEGER,          -- midpoint
  rent_estimate   INTEGER,
  monthly_mortgage INTEGER,
  monthly_cash_flow INTEGER,
  cash_on_cash_return DECIMAL(5, 2), -- e.g., 6.20
  flip_profit     INTEGER,
  mao             INTEGER,

  -- All results as JSONB (for flexibility / future fields)
  results          JSONB,

  -- Deal signals
  deal_signal_rental TEXT CHECK (deal_signal_rental IN ('green', 'yellow', 'red')),
  deal_signal_flip   TEXT CHECK (deal_signal_flip IN ('green', 'yellow', 'red')),
  deal_signal        TEXT CHECK (deal_signal IN ('green', 'yellow', 'red')),

  -- Metadata
  is_service_area BOOLEAN DEFAULT false,
  arv_method       TEXT,          -- 'comp_based' or 'zestimate_adjusted'
  rent_method      TEXT,          -- 'ratio' or 'bedroom_based'

  created_at       TIMESTAMPTZ DEFAULT now()
);

-- Indexes
CREATE INDEX idx_analyses_property ON analyses (property_id);
CREATE INDEX idx_analyses_deal_signal ON analyses (deal_signal);
CREATE INDEX idx_analyses_service_area ON analyses (is_service_area);
CREATE INDEX idx_analyses_created ON analyses (created_at DESC);
```

Why both explicit columns AND JSONB? Explicit columns let you query and filter (`WHERE deal_signal = 'green'`). JSONB stores the full payload for future-proofing without migrations.

3. `leads`

Every email captured. The CRM core.

```
CREATE TABLE leads (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),

  -- Contact info
  email       TEXT NOT NULL,
  name        TEXT,
  phone       TEXT,

  -- Linked data
  analysis_id  UUID REFERENCES analyses(id) ON DELETE SET NULL,
  property_id  UUID REFERENCES properties(id) ON DELETE SET NULL,

  -- Deal context (denormalized for CRM views – no joins needed)
  address      TEXT,
  is_service_area BOOLEAN DEFAULT false,
  deal_signal   TEXT,
  deal_arv      INTEGER,
  deal_flip_profit INTEGER,
  deal_cash_flow INTEGER,
  deal_condition TEXT,

  -- CRM fields
  status       TEXT DEFAULT 'new' CHECK (status IN (
    'new', 'contacted', 'qualified',
    'proposal_sent', 'converted', 'client',
    'nurture', 'lost'
  )),
  source       TEXT DEFAULT 'analyzer' CHECK (source IN (
    'analyzer', 'asset_group', 'content',
    'referral', 'manual'
  )),
  tags         TEXT[] DEFAULT '{}',
  notes        TEXT,

  -- Qualification
  qualification_score INTEGER,      -- 0-25 from the DEAL scorecard
  budget_range   TEXT,             -- "$50K-$100K", "$100K-$200K", etc.
  strategy       TEXT,             -- 'flip', 'hold', 'undecided'
  experience_level TEXT,           -- 'first_deal', 'beginner', 'experienced'
  financing_type  TEXT,            -- 'cash', 'hard_money', 'dscr', 'conventional'

  -- Tracking
  utm_source     TEXT,
  utm_medium     TEXT,
  utm_campaign   TEXT,

  -- Automation
```

```

email_sequence TEXT,                -- 'cook_county_flow', 'national_nurture', etc.
sequence_paused BOOLEAN DEFAULT false,

-- Timestamps
last_contacted_at TIMESTAMPTZ,
created_at      TIMESTAMPTZ DEFAULT now(),
updated_at      TIMESTAMPTZ DEFAULT now()
);

-- Indexes
CREATE UNIQUE INDEX idx_leads_email ON leads (email);
CREATE INDEX idx_leads_status ON leads (status);
CREATE INDEX idx_leads_service_area ON leads (is_service_area);
CREATE INDEX idx_leads_deal_signal ON leads (deal_signal);
CREATE INDEX idx_leads_source ON leads (source);
CREATE INDEX idx_leads_created ON leads (created_at DESC);

-- Composite index for CRM default view (service area leads with strong signals first)
CREATE INDEX idx_leads_crm_view ON leads (is_service_area DESC, deal_signal, created_at DESC);

```

4. `lead_activity`

Activity log for every interaction with a lead. Append-only — never edit or delete.

```

CREATE TABLE lead_activity (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  lead_id     UUID NOT NULL REFERENCES leads(id) ON DELETE CASCADE,

  type        TEXT NOT NULL CHECK (type IN (
    'email_sent', 'email_opened', 'email_clicked',
    'sms_sent', 'sms_replied',
    'call_completed', 'call_missed',
    'note_added', 'status_changed',
    'tag_added', 'tag_removed',
    'meeting_booked', 'meeting_completed',
    'proposal_sent', 'agreement_signed'
  )),

  notes       TEXT,                -- "Discussed Oak Forest property. Ready to move."
  metadata    JSONB,              -- Flexible payload (email subject, call duration, etc.)
  created_by  TEXT DEFAULT 'system', -- 'system', 'team_member_name', etc.
  created_at  TIMESTAMPTZ DEFAULT now()
);

-- Indexes
CREATE INDEX idx_activity_lead ON lead_activity (lead_id);
CREATE INDEX idx_activity_type ON lead_activity (type);
CREATE INDEX idx_activity_created ON lead_activity (created_at DESC);

```

5. projects

When a lead converts to a client and a rehab project begins. Ties everything together.

```
CREATE TABLE projects (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  lead_id     UUID REFERENCES leads(id) ON DELETE SET NULL,
  property_id UUID REFERENCES properties(id) ON DELETE SET NULL,
  analysis_id UUID REFERENCES analyses(id) ON DELETE SET NULL,

  -- Project info
  project_name TEXT,          -- "1234 W Chicago Ave Rehab"
  address      TEXT NOT NULL,
  strategy     TEXT CHECK (strategy IN ('flip', 'hold')),

  -- Status
  status       TEXT DEFAULT 'planning' CHECK (status IN (
    'planning', 'approved', 'in_progress',
    'punch_list', 'lease_up', 'stabilized',
    'completed', 'cancelled'
  )),

  -- Scope & Budget
  sow          JSONB,          -- Full scope of work (line items)
  condition_tier TEXT,
  budget_total INTEGER,
  budget_spent INTEGER DEFAULT 0,
  contingency   INTEGER,

  -- Timeline
  estimated_start DATE,
  estimated_end   DATE,
  actual_start    DATE,
  actual_end      DATE,

  -- Payment
  payment_schedule JSONB,      -- Array of {milestone, amount, status, paid_at}

  -- Leasing (populated during lease-up phase)
  rent_listed    INTEGER,
  rent_actual     INTEGER,
  tenant_name     TEXT,
  lease_start     DATE,
  lease_end       DATE,

  -- Outcome
  final_arv       INTEGER,
  final_rehab_cost INTEGER,
  final_cash_flow INTEGER,

  -- Timestamps
  created_at      TIMESTAMPTZ DEFAULT now(),
  updated_at      TIMESTAMPTZ DEFAULT now()
);

-- Indexes
CREATE INDEX idx_projects_lead ON projects (lead_id);
```

```
CREATE INDEX idx_projects_status ON projects (status);
CREATE INDEX idx_projects_created ON projects (created_at DESC);
```

6. `project_updates`

Weekly updates and milestone logs for active projects. Client-facing.

```
CREATE TABLE project_updates (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id  UUID NOT NULL REFERENCES projects(id) ON DELETE CASCADE,

  type        TEXT NOT NULL CHECK (type IN (
    'weekly_update', 'milestone', 'change_order',
    'inspection', 'issue', 'photo_update',
    'budget_update', 'completion'
  )),

  title       TEXT,                -- "Week 3 Update" or "Rough-In Inspection Passed"
  content     TEXT,                -- Markdown-formatted update body
  photos      TEXT[],              -- Array of photo URLs (Supabase Storage)

  -- For change orders
  change_amount INTEGER,           -- Additional cost (positive) or credit (negative)
  change_approved BOOLEAN,

  -- Budget snapshot at time of update
  budget_spent_to_date INTEGER,
  budget_remaining    INTEGER,

  created_by TEXT DEFAULT 'system',
  created_at TIMESTAMPTZ DEFAULT now()
);

-- Indexes
CREATE INDEX idx_updates_project ON project_updates (project_id);
CREATE INDEX idx_updates_type ON project_updates (type);
CREATE INDEX idx_updates_created ON project_updates (created_at DESC);
```

7. `system_defaults`

Configuration values used by the calculation engine. Editable without code changes.

```
CREATE TABLE system_defaults (
  key      TEXT PRIMARY KEY,
  value    DECIMAL(10, 4) NOT NULL,
  label    TEXT,                -- Human-readable label
  category TEXT,                -- 'finance', 'expense', 'rehab', 'deal_signal'
  updated_at TIMESTAMPTZ DEFAULT now()
```

```

);

-- Seed data
INSERT INTO system_defaults (key, value, label, category) VALUES
  ('down_payment_pct', 0.2500, 'Down Payment %', 'finance'),
  ('interest_rate', 0.0750, 'Interest Rate', 'finance'),
  ('loan_term_years', 30, 'Loan Term (years)', 'finance'),
  ('hold_months', 6, 'Flip Hold Time (months)', 'finance'),
  ('vacancy_rate', 0.0800, 'Vacancy Rate', 'expense'),
  ('mgmt_rate', 0.1000, 'Property Management Rate', 'expense'),
  ('maintenance_rate', 0.0600, 'Maintenance Rate', 'expense'),
  ('capex_rate', 0.0500, 'CapEx Reserve Rate', 'expense'),
  ('insurance_monthly', 100, 'Monthly Insurance', 'expense'),
  ('property_tax_rate', 0.0150, 'Annual Property Tax Rate', 'expense'),
  ('closing_cost_buy_pct', 0.0200, 'Buyer Closing Costs %', 'finance'),
  ('closing_cost_sell_pct', 0.0800, 'Seller Closing Costs %', 'finance'),
  ('holding_cost_monthly_pct', 0.0100, 'Monthly Holding Cost %', 'finance'),
  ('mao_arv_multiplier', 0.7000, 'MAO ARV Multiplier (70% rule)', 'finance'),
  ('default_rent_ratio', 0.0090, 'Rent-to-Value Ratio', 'finance'),
  ('rehab_cosmetic_low', 10, 'Cosmetic $/sqft (low)', 'rehab'),
  ('rehab_cosmetic_high', 20, 'Cosmetic $/sqft (high)', 'rehab'),
  ('rehab_light_low', 20, 'Light $/sqft (low)', 'rehab'),
  ('rehab_light_high', 35, 'Light $/sqft (high)', 'rehab'),
  ('rehab_medium_low', 35, 'Medium $/sqft (low)', 'rehab'),
  ('rehab_medium_high', 55, 'Medium $/sqft (high)', 'rehab'),
  ('rehab_heavy_low', 55, 'Heavy $/sqft (low)', 'rehab'),
  ('rehab_heavy_high', 85, 'Heavy $/sqft (high)', 'rehab'),
  ('rehab_gut_low', 85, 'Gut $/sqft (low)', 'rehab'),
  ('rehab_gut_high', 150, 'Gut $/sqft (high)', 'rehab'),
  ('signal_cashflow_green', 300, 'Green signal: min cash flow', 'deal_signal'),
  ('signal_cashflow_red', 0, 'Red signal: below cash flow', 'deal_signal'),
  ('signal_flip_green', 30000, 'Green signal: min flip profit', 'deal_signal'),
  ('signal_flip_red', 10000, 'Red signal: below flip profit', 'deal_signal');

```

Why a table instead of hardcoded constants? So you can tweak rehab rates, interest rates, and signal thresholds without redeploying code.

Row Level Security (RLS) Policies

```

-- Properties: public read (anyone can analyze), no public write
ALTER TABLE properties ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Public read" ON properties FOR SELECT USING (true);
CREATE POLICY "Service write" ON properties FOR INSERT
  WITH CHECK (auth.role() = 'service_role');

-- Analyses: public read (results are shareable), service write
ALTER TABLE analyses ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Public read" ON analyses FOR SELECT USING (true);
CREATE POLICY "Service write" ON analyses FOR INSERT
  WITH CHECK (auth.role() = 'service_role');

```



```
-- Leads: admin only (not public)
ALTER TABLE leads ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Admin read/write" ON leads
    USING (auth.role() = 'service_role');

-- Lead activity: admin only
ALTER TABLE lead_activity ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Admin read/write" ON lead_activity
    USING (auth.role() = 'service_role');

-- Projects: admin only
ALTER TABLE projects ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Admin read/write" ON projects
    USING (auth.role() = 'service_role');

-- Project updates: admin only
ALTER TABLE project_updates ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Admin read/write" ON project_updates
    USING (auth.role() = 'service_role');

-- System defaults: public read (frontend needs these), admin write
ALTER TABLE system_defaults ENABLE ROW LEVEL SECURITY;
CREATE POLICY "Public read" ON system_defaults FOR SELECT USING (true);
CREATE POLICY "Admin write" ON system_defaults FOR UPDATE
    USING (auth.role() = 'service_role');
```

Supabase Storage Buckets

BUCKET	PURPOSE	ACCESS
project-photos	Rehab progress photos, before/after	Private (signed URLs)
documents	SOWs, agreements, stabilization packages	Private
listing-photos	Lease-up listing photos	Public

Useful Queries

CRM Default View (Hot leads first)

```
SELECT email, address, is_service_area, deal_signal,
       deal_flip_profit, deal_cash_flow, status, source, created_at
FROM leads
WHERE status NOT IN ('lost', 'nurture')
ORDER BY is_service_area DESC,
```

```
CASE deal_signal WHEN 'green' THEN 1 WHEN 'yellow' THEN 2 ELSE 3 END,
created_at DESC;
```

Weekly Analysis Volume

```
SELECT date_trunc('week', created_at) AS week,
COUNT(*) AS total_analyses,
COUNT(*) FILTER (WHERE is_service_area = true) AS cook_county,
COUNT(*) FILTER (WHERE deal_signal = 'green') AS green_deals
FROM analyses
GROUP BY 1
ORDER BY 1 DESC;
```

Lead Conversion Funnel

```
SELECT status, COUNT(*) AS count,
ROUND(COUNT(*)::numeric / SUM(COUNT(*) OVER ()) * 100, 1) AS pct
FROM leads
GROUP BY status
ORDER BY CASE status
WHEN 'new' THEN 1 WHEN 'contacted' THEN 2 WHEN 'qualified' THEN 3
WHEN 'proposal_sent' THEN 4 WHEN 'converted' THEN 5 WHEN 'client' THEN 6
WHEN 'nurture' THEN 7 WHEN 'lost' THEN 8 END;
```

Project Budget Tracking

```
SELECT p.address, p.budget_total, p.budget_spent,
p.budget_total - p.budget_spent AS remaining,
ROUND(p.budget_spent::numeric / p.budget_total * 100, 1) AS pct_spent,
p.status
FROM projects p
WHERE p.status IN ('in_progress', 'punch_list')
ORDER BY pct_spent DESC;
```

Migration Notes

- MVP: Tables 1–4 + 7 (`properties` , `analyses` , `leads` , `lead_activity` , `system_defaults`)
- Phase 2: Add `projects` + `project_updates` when Asset Group takes first client
- Phase 3: Add user accounts table when the dashboard is built

Next Document → **10_Legal.md** — *Legal + Disclaimers*